

Week 1, part 2 of 3

Law of the instrument

"If all you have is a hammer, everything looks like a nail."

Do we really need ML?

Before you reach for a model, ask if the answer is a rule, a lookup, or a regex.

Task and model are separate choices

- **Task**: classify, rank, forecast, or generate an answer.
- **Approach**: rules, a supervised model, a pretrained model, or a combination.
- **Evidence**: labelled examples and checks matched to the task.

Before machine learning

Google's *Rules of ML* opens with three rules you apply *before* any model exists.¹

1. **Don't be afraid to launch without ML.**
2. **Design and implement metrics first.**
3. **Choose ML over a complex heuristic.**

1. Google, *Rules of Machine Learning*, "Before Machine Learning". <https://developers.google.com/machine-learning/guides/rules-of-ml> ↩

Rule #1: Don't be afraid to launch without ML

"Machine learning is cool, but it requires data."¹

- A simple heuristic gives you a baseline before investing in a trained model.
- Rank apps by install rate. Block the publishers you already know are spam. Curate by hand.
- Without task data, you cannot evaluate reliably or train a bespoke model. A pretrained model may still be useful.

1. Google, *Rules of Machine Learning*, Rule #1. <https://developers.google.com/machine-learning/guides/rules-of-ml> ↩

Rule #2: Design and implement metrics first

"Before formalizing what your machine learning system will do, track as much as possible."¹

- Collect only the evidence you need, with appropriate consent and access controls.
- Historical data is only available if you started logging before you needed it.
- Instrumenting metrics shapes the system you end up building.
- You learn what actually moves, versus what you assumed would.

1. Google, *Rules of Machine Learning*, Rule #2. <https://developers.google.com/machine-learning/guides/rules-of-ml> ↩

Rule #3: Choose ML over a complex heuristic

"A complex heuristic is unmaintainable."¹

Simple rules ship your first version. But once you have data and a clear objective, and the rules have grown into a thicket of special cases, that is the signal to switch. Models are easier to update and maintain than a hundred hand-tuned conditionals.

1. Google, *Rules of Machine Learning*, Rule #3. <https://developers.google.com/machine-learning/guides/rules-of-ml> ↩

Choose the intervention from failure

What is missing?	Candidate approach
A stable rule or lookup	Ordinary code
A label, score or ranking	Majority baseline, then supervised ML
Task instructions or examples	Prompt a pretrained model
Evidence from changing documents	Retrieval
A permitted action or calculation	A tool, with runtime checks
Consistent task behaviour from labelled examples	Fine-tuning

RECAP: LAW OF THE INSTRUMENT

1. Ask whether a rule, a lookup, or a regex answers it first.
2. Choose the task, baseline and evaluation separately from how the model was trained.
3. Launch without ML to earn the data. Switch when the heuristic stops being maintainable.
4. Design your metrics before the system, not after it.
5. Choose an intervention that addresses the observed failure.

NEXT: YOU HAVE DECIDED YOU NEED
A MODEL. HOW DO YOU DECIDE?